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Microaerobic bio-pretreatment of straw with fermentation broth as inoculum: Enhancing hydrolysis-acidification efficiency

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ABSTRACT

Anaerobic digestion (AD) has been recognized as a promising technology for the disposal of agricultural residues, such as straw. Nonetheless, AD encounters a bottleneck of low hydrolysis rate. Microaerobic pretreatment significantly enhances straw AD efficiency, but its effectiveness depends on factors like pretreatment time and water content, with their interactions remaining unclear. Therefore, this study employed a response surface methodology (RSM) to investigate the interactions among pretreatment time, water content, and volatile fatty acids (VFAs) concentration in inoculum during the microaerobic pretreatment. The results showed that the highest hydrolysis-acidification efficiency was achieved with a pretreatment time of 10 days and a VFAs concentration of 10,000 mg/L, and there was a notable interaction between these two factors in the inoculum. Additionally, microbial community analysis showed that microaerobic pretreatment altered the microbial community structure to adapt to micro-oxygen environments, promoting the hydrolysis-acidification of straw. Specifically, *Proteobacteria*, *Firmicutes*, and *Actinobacteria* became the predominant acid-producing bacteria following microaerobic pretreatment. Within the fungus communities, *Aspergillus niger* and *Penicillium citrinum*, known for lignin degradation, increased by 4.4-fold and 1.2-fold, respectively. And the *Methanobacteria* became the main group in the archaeal communities. This paper presents a new perspective on the pretreatment of lignocellulose in commercial anaerobic reactors.

1. Introduction

With an annual production of 1.04 billion tons, straw stands as one of the primary agricultural wastes in China [1]. The disposal of such a substantial amount of waste has posed a longstanding challenge in the realm of energy and environmental management. Anaerobic digestion (AD) is an advantageous technology for converting straw into methane, offering the potential to simultaneously tackle environmental issues [2]. However, the primary constituent of straw, lignocellulose, comprises a intricate structure that is intertwined with lignin, cellulose, and hemicellulose, thereby impeding efficient degradation by hydrolytically acidifying bacteria [3,4]. Consequently, overcoming the rate-limiting step (hydrolysis) [5], and converting lignocellulosic biomass into soluble substances are crucial for enhancing the AD process.

To address these challenges, recent advancements in biological, chemical, mechanical, and thermal pretreatments have been recognized as effective methods for achieving high substrate solubilization[6–8]. Among these methods, biological pretreatment has garnered considerable attention due to its mild reaction conditions and low cost. Notably, microaerobic biological pretreatment demands only minimal amount of oxygen, making it an economically efficient and environmentally friendly alternative pretreatment option [9].Recent evidence indicates that the introduction of small quantities of oxygen could improve the relative abundance and bacterial diversity of *Firmicutes*, thereby increasing the hydrolysis rate of straw [10,11]. However, this approach entails not only the cost of inoculants but also requires the supplementation of nutrients such as nitrogen and trace elements [12]. In addition, microbial pretreatment requires the additional addition of strains or

Abbreviations: AD, Anaerobic digestion; RSM, response surface methodology; VFAs, volatile fatty acids; TS, total solids; VS, volatile solids; SCOD, soluble chemical oxygen demand.

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hydrolytic enzymes that can promote straw hydrolysis. The selection of such strains [13] and the high cost of purchasing related hydrolytic enzymes pose significant barriers to the widespread application of microbial pretreatment (e.g., enzyme treatment [14] and white-rot fungus treatment [15]).

Meanwhile, the fermentation broth generated from the hydrolysis-acidification reactor contains abundant microbial populations and nutrient. Previous research has found that the fermentation broth contains 16 bacterial phyla and 7 fungus classes, along with a high content of ammonia nitrogen [16]. Previous studies have shown that using biogas slurry for straw pretreatment is feasible [17], as it contains high concentrations of ammonium nitrogen and organic acids that can dissolve lignocellulose, thereby promoting biogas production [18]. In addition, aerobic composting techniques also involve stacking organic materials such as straw and spraying biogas slurry for pretreatment. Past research demonstrated that organic acids were promoted subcritical pretreatment effectively, increased the yield of polysaccharide and inhibited its degradation to furfural [19,20]. Organic acids enriched in Volatile fatty acids (VFAs) can release soluble sugars during the pretreatment process, thereby increasing the hydrolysis reaction rate [21]. In addition, there are various influencing factors in pretreatment [22], for instance, pretreatment time [23] and water content. Longer pretreatment times resulted in a loss of organics, while shorter pretreatment times led to inadequate decomposition of substrate [24].VFAs as organic acids, a crucial constituent within the fermentation broth, however, past studies lack systematic research on their relationship with factors such as water content and pretreatment time during the pretreatment process. Consequently, it is essential to thoroughly investigate the combined impacts of these influencing factors in fermentation broth pretreatment and further optimize the specialized condition to enhance the efficiency of pretreatment.

Experimental designs frequently employ response surface methodology (RSM), which effectively integrates mathematical and statistical techniques [25,26]. It is the capability of RSM to extract important parameters from multiple factors for analyzing the interaction effects of dependent variables in experiments [27]. Previous study has utilized RSM to optimize thermal hydrolysis pretreatment of sewage sludge [28]. By applying RSM, it is possible to identify the optimal temperature and reaction time, as well as analyze the interactions among other factors affecting straw hydrolysis efficiency [29]. Therefore, utilizing RSM could adequately determine the optimum conditions in the hydrolysis-acidification of straw.

Moreover, the key to overcoming the rate-limiting step lies in enhancing the presence of microorganisms capable of decomposing lignocellulosic materials, including *Actinobacteria*, *Firmicutes*, and *Proteobacteria* during pretreatment [30]. Previous study has indicated limited oxygen promote facultative anaerobic microflora associated with hydrolysis-acidification of lignocellulosic [31]. To date, it is generally recognized that pretreatment conditions significantly influence the effect of pretreatment microorganism [20]. However, the investigation of microbial community changes in the pretreatment of hydrolytic fermentation liquid as inoculum under microaerobic conditions remains unclear.

This study investigated the effects and interactions of pretreatment time, concentration of VFAs in inoculum, and water content on the hydrolysis and acidification efficiency of straw under microaerobic conditions using fermentation liquid as the inoculum, employing RSM. The aim to better optimize the specific conditions — specifically, pretreatment time, VFAs concentration in inoculum, and water content — for straw hydrolysis and acidification to enhance conversion efficiency. In addition, high-throughput sequencing was chosen to investigate the impacts of microaerobic pretreatment application in AD on the microbial community process, and to explore the microbiological mechanisms involved during the pretreatment process. This study aims to provide practical strategies for improving the precision control and efficiency of straw pretreatment.

2. Experimental

2.1. Substrate and inoculum

The rape straw used for pretreatment was collected and subjected to natural air drying in an open-air yard until the water content reached below 10 %. After removing any remaining grain residues from the straw, it was mechanically crushed and the resulting particles were sifted using a stainless steel screen with a 20-mesh size, retaining particles larger than 30-mesh. The straw exhibited carbon (C) and nitrogen (N) contents of 44.51 ± 0.32 % and 0.88 ± 0.11 %, respectively. The total solids (TS) and volatile solids (VS) of straw were 91.21 ± 0.66 % and 86.63 ± 0.53 %, respectively.

The inoculum was fermentation broth which obtained from the hydrolysis-acidification reactor (80 L) of a laboratory-scale two-phase anaerobic reactor during its normal operation (with TS and VS of 6.68 % and 2.93 %, respectively). The inoculum exhibited a pH of 6.81 ± 0.16, accompanied by concentrations of soluble chemical oxygen demand (sCOD), volatile fatty acid (VFAs), and ammonia nitrogen at 0.78 ± 0.11 g/L, 0.45 ± 0.08 g/L, and 3.2 ± 0.18 g/L, respectively.

2.2. Experimental design

2.2.1. Microaerobic pretreatment

In order to determine the optimal oxygen load for micro-aerobic pretreatment, different oxygen loads were applied (0, 20, 60, 120 mL/g VS/d). The specific operation is as follows: In a wide-mouth bottle (500 mL), 35 g of rape straw (based on TS) and 58 mL inoculum were introduced, with a microaerobic pretreatment duration of 0–20 days. The pretreatment temperature was set at 25℃. The wide-mouth bottle had a sealed rubber stopper with dual perforations through which polyethylene tubes were inserted. An aeration head was placed at the bottom of the bottle, and a micro air pump (1 L/min) supplied air into a polyethylene tube inserted at the bottom of the bottle. The optimal conditions for this experiment were established based on the effectiveness of lignin degradation. Through this experiment, ensure that the oxygen load of pretreatment was maintained at 20 mL/g VS/d, which corresponds to the optimal microaerobic conditions.

2.2.2. Hydrolysis-acidification and response surface experiments

Pre-experiments under microaerobic were carried out to assess the impact of pretreatment time, concentration of VFAs in inoculum, and water content on hydrolysis-acidification efficiency of straw. Each experiment involved placing 12 g of pretreated straw (based on TS) in a 250 mL bottle. The fermentation broth from a two-phase anaerobic reactor was added as the inoculum. The effective volume was 200 mL and the TS loading rate was 60 g TS/L. The bottle was sealed with an aluminum-plastic cap, thoroughly mixed, and then placed in a shaking water bath set at constant temperature (25℃), rotating at 100 r/min. Afterwards, the fermentation liquid in the bottle was collected daily to analyze the changes in VFAs concentrations (final concentrations were calculated by subtracting VFAs in the inoculum) [16]. Through pre-experiments, the effects of individual factors on the hydrolysis and acidification efficiency of straw were obtained.

Based on the results of single-factor experiments, a systematic analysis of the impacts of these three factors can better optimize the straw hydrolysis and acidification process, thereby improving conversion efficiency. A Box-Behnken response surface design with three factors at three levels were conducted to obtain the optimal hydrolysis-acidification efficiency of straw. The impact of altering one factor on the results was investigated by holding two out of the three factors constant. In this study, pretreatment time was chosen as x_1 (7, 10, and 13 d), concentration of VFAs in inoculum was chosen as x_2 (5000, 10000, and 15000 mg/L), and water content was chosen as x_3 (60 %, 65 %, and 70 %). The variation in VFA concentration was achieved by adjusting the proportions of each organic acid in the inoculum,

specifically, acetic acid, propionic acid, butyric acid, and pentanoic acid were present in the total VFAs in the ratios of 60 %, 15 %, 20 %, and 5 %, respectively. The VFAs concentration of straw hydrolysis-acidification was set as the response variable(Y). A total of 17 groups were designed for the B-B response surface experiment. The experimental operating steps are the same as those for the pre-experiments. All experiments were conducted under microaerobic conditions (20 mL/g VS/d oxygen load). In the calculation of VFAs concentration during the hydrolysis and acid production of the straw, the amount of VFAs introduced with the inoculum has already been subtracted.

2.3. Microbial analysis

To explore alterations in the microbial community following microaerobic pretreatment, the inoculum of pretreatment and the straw samples after pretreatment were stored at −20°C and labeled as 'before' and 'after', respectively. 100 mL of the inoculum sample was filtered through a 0.22 μm pore-size membrane to retain solid samples from the inoculum for subsequent analysis. 20 g of straw sample was collected at the end of pretreatment under optimal micro-oxygen conditions (20 mL/g VS/d oxygen load, 65 % water content, and pretreatment temperature of 25°C). To compare and analyze variations in bacterial, archaeal, and fungus community structures between the 'before' and 'after' groups, we utilized high-throughput sequencing.

DNA extraction from the samples was carried out using a soil DNA extraction kit (Omega Bio-tek, Norcross, GA, USA. The V4 hypervariable region of the 16S rRNA gene and the ITS1 region were amplified by PCR separately. The PCR protocol included an initial denaturation at 95°C for 5 min, followed by denaturation, annealing, and extension cycles, and a final extension step. PCR reactions were conducted in triplicate to ensure accuracy in the results.

The primer sequences used for the V4 hypervariable region were 515 F 5'-GTGCCAGCMGCCFCFF-3' and 806 R 5'-GGACTACHVGGGTWTCTAAT-3', while for the ITS region, the primer sequences were its1–5'-TCCGTAGGTGAACCTGCGG-3' and its4–5'-TCCCTCCGCTTATTGATATGC-3'. To ensure experimental accuracy, PCR reactions were conducted in triplicate to ensure accuracy in the results.

2.4. Data analysis methods

The original data were processed using the calculation functions in Microsoft Office Excel, which is the 2016 version released by Microsoft. All figures were created using Origin 2021.

3. Result and discussion

3.1. Analyzing interactions in microaerobic pretreatment with RSM

The VFAs results of different combination of the pretreatment time, VFAs concentration in inoculum, and water content were shown in the Table S1. Based on this, VFAs concentration was chosen as the output response, and a quadratic response surface regression model was established for the pretreatment time, VFAs in inoculum and water content during the pretreatment process. The quadratic Eq. (1) was generated by the software:

$$y = -96737.27356 + 1606.82403\times_1 + 0.584056\times_2 + 2855.27102\times_3 - 0.015851x_1x_2 - 2.45217x_1x_3 + 0.009549x_2x_3 - 65.35547\times_1^2 - 0.000048\times_2^2 - 22.78317\times_3^2 \quad (1)$$

In this context, y represents the VFAs concentration, while x1, x2, and x3 correspond to pretreatment time (d), concentration of VFAs in inoculum (mg/L), and water content (%), respectively. Additionally, Table 1 showed the outcomes of the analysis of variance (ANOVA).

The 'Prob > F' values were below 0.0001, with value below 0.05 indicate that the strong significance of the model and its terms, including x_2, x_3, $x_1\times_2$, $x_2\times_3$, x_1^2, x_2^2, and x_3^2. The coefficient of determination ($R^2 = 0.9802$) implied that the model is highly effective, capable of explaining up to 98.02 % of the response variability. Therefore, these terms could not be removed to maintain the integrity of the model hierarchy, despite the insignificance of the model term $x_1\times_3$ ($p > 0.05$).

Table 1
ANOVA results for VFAs concentration.

Source	Sum of Squares	F-value	P-value Prob > F
Model	1.217E+ 07	38.50	< 0.0001
x_1	23804.53	0.6777	0.4375
x_2	1.802E+ 06	51.31	0.0002
x_3	2.531E+ 05	7.21	0.0313
$x_1\times_2$	2.261E+ 05	6.44	0.0388
$x_1\times_3$	5411.81	0.1541	0.7064
$x_2\times_3$	2.279E+ 05	6.49	0.0382
x_1^2	1.457E+ 06	41.47	0.0004
x_2^2	5.953E+ 06	169.50	< 0.0001
x_3^2	1.366E+ 06	38.89	0.0004

The interaction effects of pretreatment time and VFAs concentration in inoculum on the hydrolysis-acidification efficiency of straw are presented in the form of contour plots and 3D surface plots in Fig. 1. As shown in Fig. 1a, the hydrolysis-acidification efficiency of straw increased with increasing pretreatment time and VFAs concentration in inoculum. The highest VFAs production achieved during straw hydrolysis-acidification was 6492.42 mg/L, with pretreatment time of 10 d and VFAs concentration in inoculum was 10000 mg/L. Beyond these values, the hydrolysis-acidification efficiency of straw decreased. For instance, at a VFAs concentration of 15000 mg/L in inoculum and a pretreatment time of 13 days, the VFAs concentration produced by straw hydrolysis-acidification significantly dropped to only 4492.04 mg/L, representing a 28.62 % decrease from the peak recorded value.

The elliptical shape of the contour lines indicated a significant interaction between pretreatment time and VFAs concentration in inoculum on the hydrolysis-acidification efficiency of straw (Fig. 1a). This is consistent with results from previous studies, which showed that the more elliptical the contour shape, the greater the interaction between the two factors. In previous studies, the two factors that displayed circular contours, the liquid-to-solid ratio and hydrolysis time, also showed a relatively small interaction [32]. In comparison to pretreatment time, the impact of VFAs concentration in inoculum on the results was more significant. For instance, in Fig. 1a, the long axis of the ellipse is aligned with the y-axis (representing VFAs concentration in inoculum), and the contour density is greater in this direction compared to the x-axis [33].

As shown in Fig. 1b, the effect surface curve showed a strong inclined trend when VFAs concentration in inoculum between 5000 mg/L to 10000 mg/L, while the inclined trend decelerated within the concentration range of 10000 mg/L to 15000 mg/L. These results indicated that hydrolysis-acidification efficiency of straw increased and then decreased as a function of VFAs concentration in inoculum. An appropriate concentration of VFAs can enhance the bioavailability of straw; however, excessively high concentrations of VFAs generated during the pretreatment process can lead to acid inhibition, which is detrimental to microbial metabolism [34]. In comparison to the effect surface related to VFAs concentration in inoculum, the effect surface associated with pretreatment time appeared flatter, indicating that the influence of pretreatment time on the production of VFAs via straw hydrolysis-acidification is limited (Fig. 1b).

These results may be related to fermentation broth as inoculum, which had been shown to be conducive to straw hydrolysis-acidification. Another possible explanation is that the combined effect of microaerobic with VFAs concentration of inoculum was higher than that with the pretreatment time. Previous research has found that the oxygen

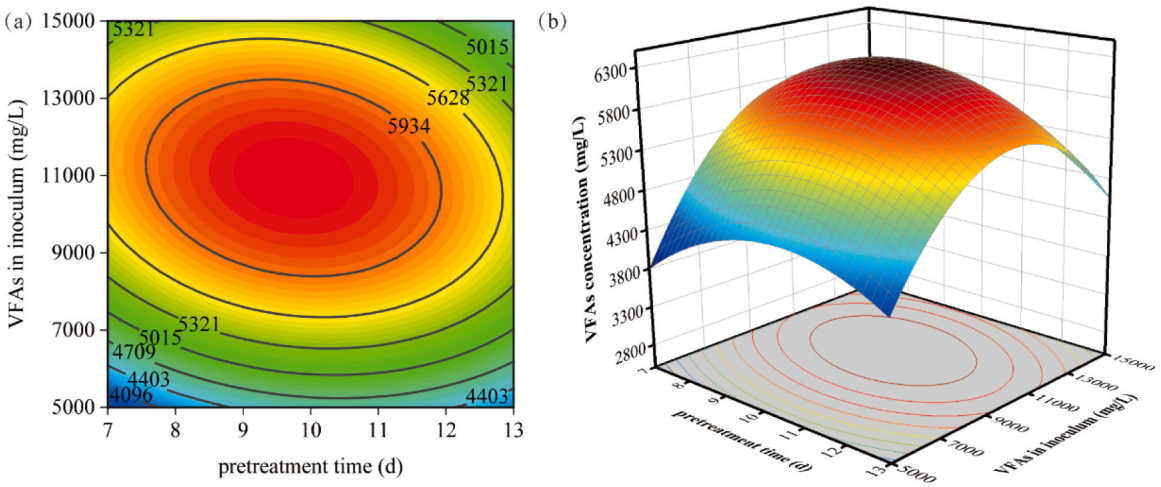


Fig. 1. The interaction between pretreatment time and volatile fatty acids (VFAs) concentration in inoculum on the hydrolysis-acidification efficiency of straw.

provided by microaerobic can promote the growth of facultative microorganisms such as *Firmicutes* and *Clostridium*, and may alter their metabolic pathways, enabling them to degrade organic matter through aerobic processes, thereby facilitating the rapid hydrolysis of substrates [35].

The interaction between VFAs concentration in inoculum and water content of substrate on the hydrolysis-acidification efficiency of straw was similar to the results analyzed in Fig. 1, with the interaction also being evident. As shown in Fig. 2a, the long axis of the ellipse extends along the x-axis (VFAs in inoculum) and the contour density is greater. Compared to water content, the effect surface of VFAs concentration in inoculum showed a steeper inclination. This indicated that VFAs concentration in inoculum had a greater impact on hydrolysis-acidification efficiency of straw. The highest hydrolysis-acidification concentration of straw, with simulated value was 6492.42 mg/L, was achieved when VFAs concentration in inoculum and water content reached 10000 mg/L and 65 %, respectively (Fig. 2b). However, exceeding these values resulted in a decrease in straw acidification efficiency. For instance, when the VFAs concentration in inoculum was 15000 mg/L and water content was 70 %, the VFAs concentration produced by straw hydrolysis-acidification was only 5161.46 mg/L. This value was 20.5 % lower than the highest recorded concentration. The contour lines are elliptical, indicating that VFA concentration in inoculum had a more significant impact on the results than water content (Fig. 2a).

The interaction effects of pretreatment time and water content on hydrolysis-acidification efficiency of straw were presented in the form of contour plots and 3D surface plots, respectively (Fig. 3). Hydrolysis-acidification efficiency of straw increased with the increase of pretreatment time and water content (Fig. 3a). The highest concentration of VFAs for straw hydrolysis-acidification, with a simulated value of 6492.42 mg/L, was achieved when pretreatment time and water content were 10d and 65 %, respectively. When it exceeds these values, the efficiency of straw acidification decreased. For example, when the pretreatment time was 13d and water content of straw was 70 %, VFAs concentration produced by straw hydrolysis-acidification was only 4824.14 mg/L, which decreased by 25.70 % compared with the highest value.

As shown in Fig. 3a, the circular contour lines indicated that pretreatment time and water content had an insignificant interaction effect on the hydrolysis-acidification efficiency of straw [32]. Fig. 3b illustrates that the general trend of the effect surfaces for the two factors was very similar and both exhibited a slight slope, suggesting that pretreatment time and water content had insignificant effects on hydrolysis-acidification efficiency of straw.

3.2. Changes in microorganisms before and after pretreatment

3.2.1. Changes in bacterial community

At the phylum level, the dominant bacterial communities underwent significant changes after pretreatment, as shown in Fig. 4a. In the

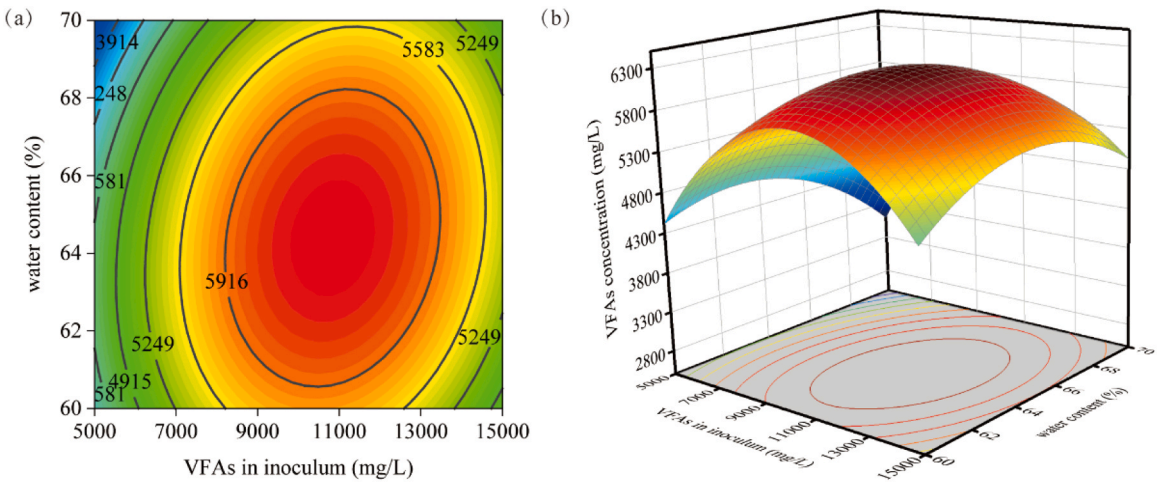


Fig. 2. The interaction between volatile fatty acids (VFAs) concentration in inoculum and water content on the hydrolysis-acidification efficiency of straw.

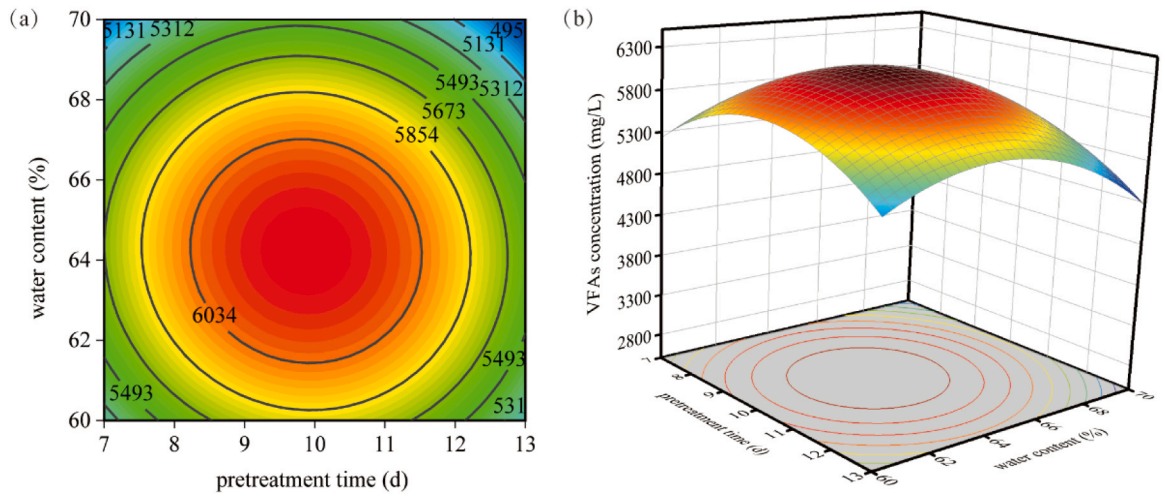


Fig. 3. The interaction between pretreatment time and water content on the hydrolysis-acidification efficiency of straw.

'before', the bacteria were mainly distributed in 16 phyla, and the major groups were *Bacteroidetes*, *Firmicutes* and *Proteobacteria*, with relative abundances of 50.26 %, 15.37 % and 9.34 %, respectively. *Bacteroidetes* were dominant, and they typically contributed to the degradation of organic substances like carbohydrates and polysaccharides, generating VFAs during AD at medium temperatures [36]. The phylum *Firmicutes* is the second most abundant bacterial phylum. And they were also an important bacterial group in anaerobic hydrolysis-acidification [37]. Additionally, *Proteobacteria* had the ability to hydrolyze and produce VFAs [38]. Since the inoculum was taken from the end stage of hydrolysis in the hydrolysis-acidification reactor, which was just prior to entering the acidogenesis and methanogenesis stages of AD, *Bacteroidetes* exhibited a higher relative abundance compared to *Firmicutes* and *Proteobacteria*, indicating their predominant involvement in the second and third stages of AD. These results seem to be consistent with other research which found *Proteobacteria*, *Firmicutes*, and *Bacteroidetes* are capable of efficiently degrading rice straw to generate organic acids [39].

Compared with 'before', the bacterial population exhibited significantly more diversity after microaerobic pretreatment. For example, *Proteobacteria*, *Firmicutes*, *Bacteroidetes*, and *Actinobacteria* were dominant, with relative abundances of 59.65 %, 23.86 %, 10.87 %, and 4.06 %, respectively. Specifically, the relative abundance of *Proteobacteria*, *Firmicutes*, and *Actinobacteria* increased by 5.3, 0.6, and 84.5 times, respectively, compared to their abundances in 'before'. It seems possible that these results are due to the presence of more oxygen promotes the proliferation and metabolic activity of facultative bacteria compared to anaerobic environment [40]. The relative abundance of *Bacteroidetes* declined by a factor of 3.6, which may be attributed to its status as an anaerobic bacterial group that was not well adapted to microaerobic conditions [30]. *Proteobacteria* and *Firmicutes* exhibited an increased relative abundance in the 'after', indicating their capability to produce more extracellular enzymes. Previous studies demonstrated that the extracellular enzymes produced by *Proteobacteria* and *Firmicutes*, such as cellulases, lipases, and proteases, played a role in promoting the degradation of macromolecular organic compounds such as cellulose, proteins, and lipids [41]. This offers a possible explanation for the improved efficiency of straw hydrolysis-acidification during AD with microaerobic pretreatment. In addition, *Actinobacteria* is one of the most potent bacterial species in degrading lignocellulose, which can use hyphae to penetrate into the inside of straw and increase the water solubility of lignocellulose [42].

At the class level, there were significant changes in the bacterial community after pretreatment (Fig. 4a). For example, the relative abundance of *Alphaproteobacteria*, *Betaproteobacteria*, and *Bacillus* was 27.56 %, 26.51 % and15.62 %, respectively, representing increases of 25.30 %, 26.12 %, and 15.57 % compared to the 'before'. This indicates that *Alphaproteobacteria* and *Betaproteobacteria*, belonging to the *Proteobacteria*, became the dominant bacterial groups in the reactor after microaerobic pretreatment. This may be due to microaerobic pretreatment disrupting the complex structure of straw, converting the complex carbon source into simpler carbon compounds derived from lignocellulose [43]. The bacteria in *Proteobacteria* have the strongest ability to deconstruct the lignin structure in cell walls [44]. Specially, the microaerobic environment provided them with growth opportunities, which will be one of the reasons for the increased hydrolysis-acidification efficiency of straw in 'after'. Additionally, *Bacillus* has also been proven to be effective in straw pretreatment [45].

In contrast, the relative abundance of *Bacteroidia* decreased significantly by 26.09 % compared to before pretreatment (Fig. 4b). This finding aligns with an earlier study that demonstrated *Bacteroidia* as the main group in acid-producing anaerobic fermentation [46], indicating weakened activity of acid-producing bacteria during anaerobic fermentation. *Clostridia*, which plays a key role in anaerobic gas production [10], also decreased by 7.35 %. After pretreatment, the relative abundance of several lignin-degrading bacteria within the *Proteobacteria*, such as members of the *Xanthomonadales* order and *Flavobacterium*, increased significantly by 9.6-fold and 21.6-fold, respectively (Fig. 4c). This demonstrates that microaerobic pretreatment could enhance the growth and metabolism of *Proteobacteria*, *Firmicutes*, and *Actinobacteria*, which was favorable for the production of hydrolytic enzymes. It also implies that microaerobic pretreatment leads to higher efficiency of straw hydrolysis-acidification.

3.2.2. Changes of archaeal community

The liquid inoculum used in this study was obtained from the hydrolysis-acidification reactor of a two-phase anaerobic reactor, which was strictly maintained under anaerobic conditions and harbored methanogens. The introduction of oxygen during the AD is toxic to methanogens [47]. However, several reports have shown that certain methanogens, specifically those belonging to the genera *Methanosarcina* and *Methanococcus*, display a tolerance to trace amounts of oxygen [10, 35].

The identified archaea, both before and after pretreatment, belong to the phylum *Euryarchaeota*. Before pretreatment, *Methanomicrobia* exhibited the highest relative abundance at the class level, specifically 87.48 %. However, following microaerobic pretreatment, a notable shift occurred in the dominant archaeal community. *Methanobacteria* became the main group, with a relative abundance of 66.67 %, representing a 65.05 % increase compared to the 'before' (Fig. 5a). As shown in Fig. 5b,

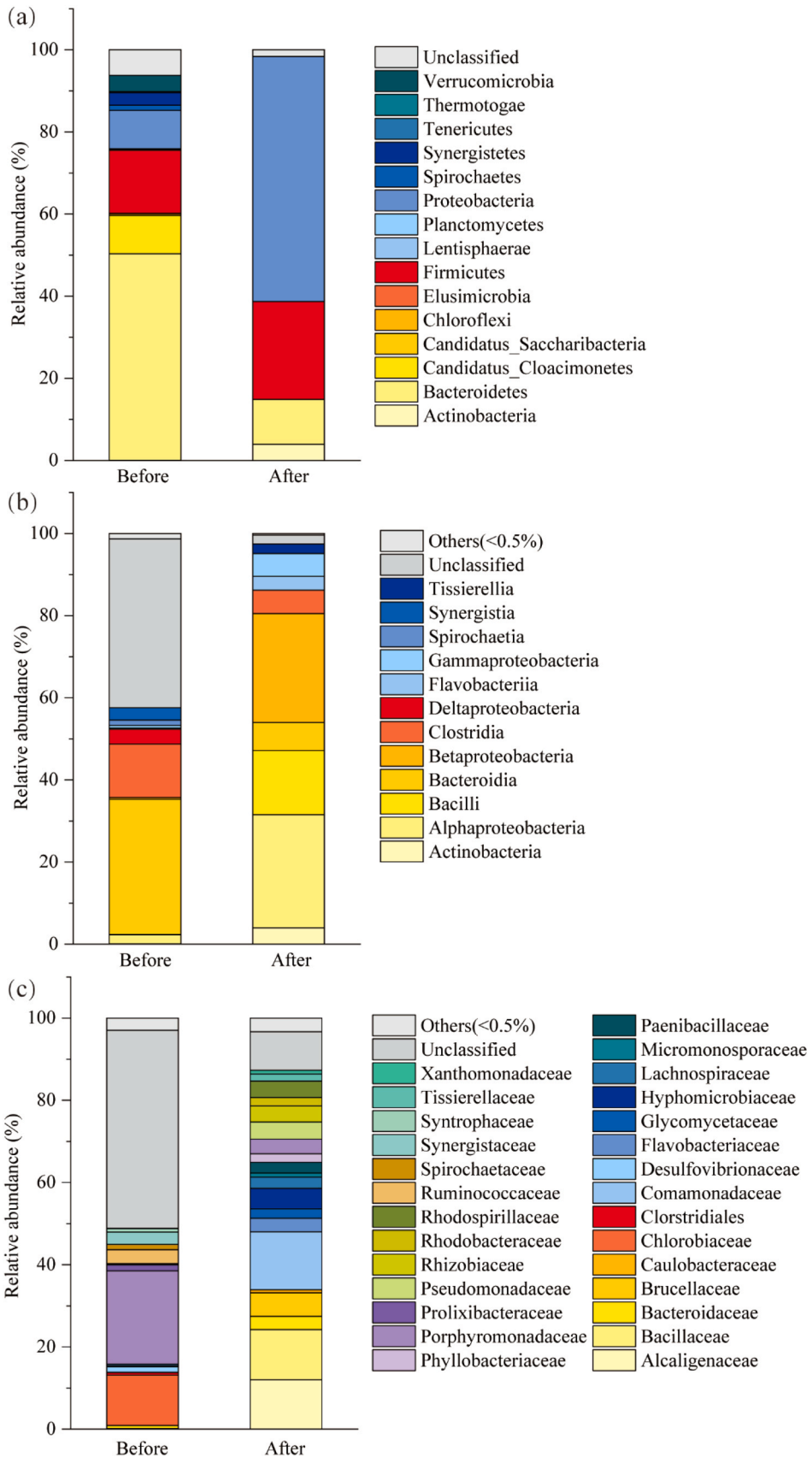


Fig. 4. Bacterial communities at different taxonomic levels in the before (inner circle) and after (outer circle) pretreatment: (a) Bacterial phylum; (b) Bacterial class; (c) Bacterial family.

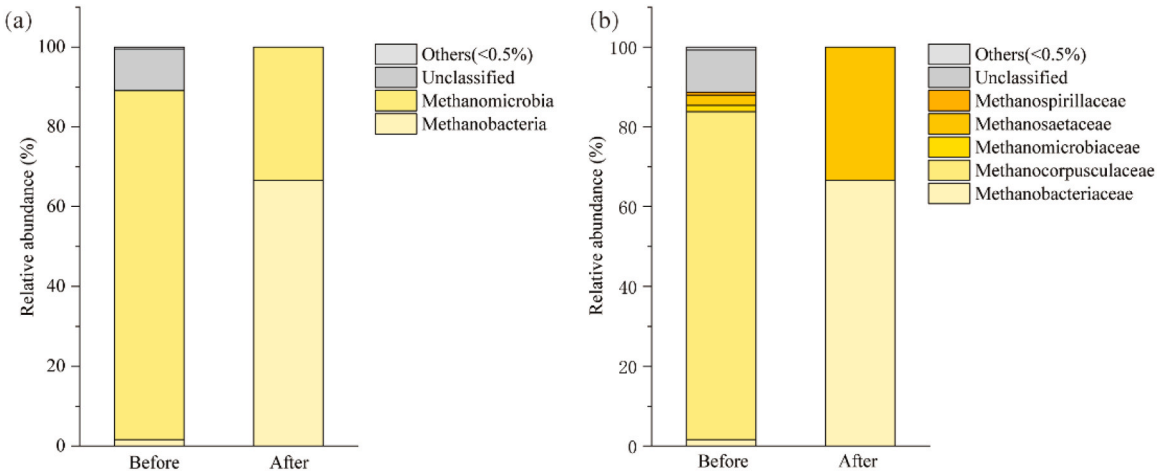


Fig. 5. Archaeal communities at different taxonomic levels in the before (inner circle) and after (outer circle) pretreatment: (a) Archaeal phylum, (b) Archaeal family.

the distribution of archaeal communities at the family level showed significant differences before and after pretreatment. In the 'before', five methanogenic archaeal families were identified, among which the family with the main group was *Methanobacteriaceae*, with a relative abundance of 82.18 %. Contrastingly, after the microaerobic pretreatment, only one family, *Methanosaetaceae*, was detected in the 'after' sample. *Methanobacteriaceae* and *Methanosaetaceae* are both common AD microorganisms. *Methanobacteriaceae* predominantly participate in the hydrogenotrophic methanogenesis pathway [48], converting hydrogen and carbon dioxide produced during the acidogenic phase into methane. In contrast, *Methanosaetaceae* primarily utilize acetate for methane production [49]. This notable shift could be attributed to microaerobic conditions that promote the acidification of straw, thereby inhibiting the proliferation of hydrogenotrophic methanogenic archaea [50]. Microaerobic pretreatment induce changes in the archaeal community structure, enabling adaptation to microaerobic conditions, leading to a significant reduction in the species of archaeal communities after pretreatment.

3.2.3. *Changes of fungus community*

Fungus are recognized for producing a variety of enzymes that degrade cellulose, hemicellulose, and lignin, and are generally regarded as more efficient in lignin degradation compared to other microorganisms, including actinomycetes and bacteria [51].

As shown in Fig. 6, the fungi community exhibited a significant change. Initially, the fungi community was mainly distributed in seven classes and sixteen species. However, after microaerobic pretreatment, the diversity of the fungi community significantly decreased, with distribution limited to two classes and three species (Fig. 6a). The dominant species with relatively high abundance after pretreatment were *Pseudallescheria boydii*, *Aspergillus niger*, and *Penicillium citrinum*, with relative abundances of 10.62 %, 7.33 %, and 1.12 %, respectively (Fig. 6b). Previous studies have confirmed that *Aspergillu* and *Penicillium* can degrade lignocellulose [52]. Moreover, the relative abundance of *Aspergillus niger* and *Penicillium citrinum*, which are associated with lignin degradation, increased by 4.4-fold and 1.2-fold, respectively, after pretreatment. This suggests that pretreatment, which primarily utilized rapeseed straw as the main carbon source, exerted a selective effect on the fungus. In contrast, some studies have shown that 0.1 %–5 % PS pretreatments have no discernible impact on the soil bacterial and fungi community [53]. In addition, the co-digestion of food waste and straw indicates a significant increase in fungus communities, promoting cooperation between fungi and bacteria, which enhances the hydrolysis efficiency of straw [54]. Therefore, using microaerobic pretreatment can significantly increase the abundance of lignin-degrading fungi, thus enhancing the lignin degradation efficiency in a cost-effective manner.

4. Conclusion

The interaction between pretreatment time and VFAs concentration in the inoculum was found to be the most significant through RSM analysis. However, the interaction between water content and

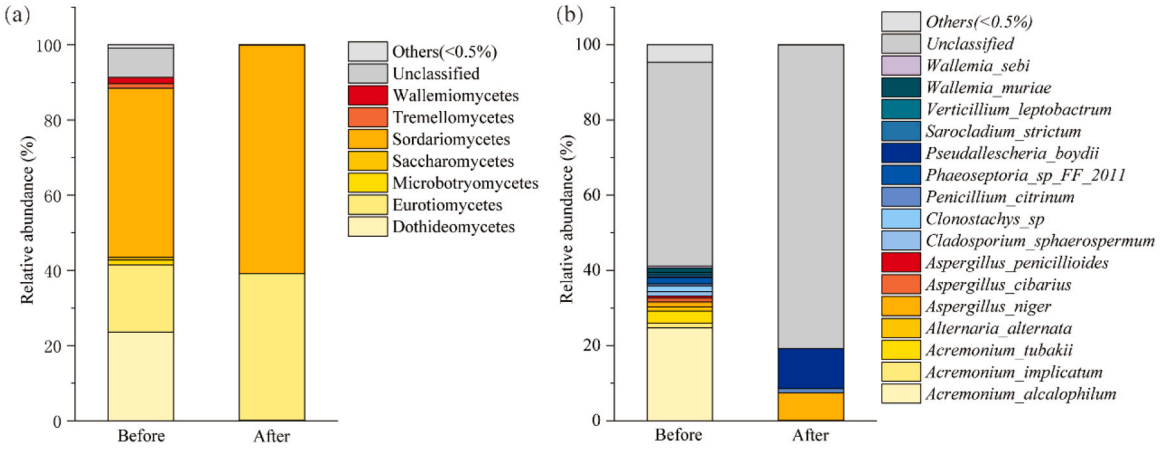


Fig. 6. Fungus communities at different taxonomic levels in the before (inner circle) and after (outer circle) pretreatment: (A) Fungus order; (B) Fungus species.

pretreatment time on straw hydrolysis-acidification was found to be relatively insignificant. The microbial community structure changed significantly after microaerobic pretreatment. The dominant bacterial community consisted mainly of *Proteobacteria*, which included a diverse range of bacteria capable of lignin degradation, replacing the dominant *Bacteroidetes* before pretreatment. Specifically, the relative abundance of *Actinobacteria*, known for their robust lignin degradation capability, increased by 84.5 times after the pretreatment. Due to oxygen presence, the archaeal community experienced a notable decrease in diversity after pretreatment, reducing from five methanogenic species before pretreatment to only *Methanosaetaceae*. In the fungi community, the relative abundance of *Aspergillus niger* and *Penicillium citrinum*, which are associated with lignin degradation, increased by 4.4-fold and 1.2-fold, respectively, after microaerobic pretreatment. This research highlights the potential of microaerobic biological pretreatment to optimize the processing of lignocellulosic materials, such as rape straw. These findings provide a basis for further practical applications, enhancing both sustainability and economic viability.

CRediT authorship contribution statement

Chuqiao Wang: Writing – original draft, Resources, Investigation, Funding acquisition, Conceptualization. **Shan Huang:** Project administration, Funding acquisition. **Xizi Long:** Supervision, Funding acquisition. **Xiaoqian Liu:** Visualization, Software, Investigation, Data curation. **Jialong Pan:** Formal analysis. **Yuying Hu:** Methodology. **Zhang Jia jie:** Writing – review & editing, Project administration, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.procbio.2026.01.005.

Data availability

Data will be made available on request.

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